Huawei Access Node KR-ACC-15 QoS Congestion Event Troubleshooting SOP

Node: KR-ACC-15 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_4295

Description

This Standard Operating Procedure outlines the steps to be taken in response to a critical QoS congestion event on Huawei Access Node KR-ACC-15, identified by Alarm Code ALARM_4295. The underlying protocol mechanics that lead to this failure domain are rooted in the interaction between the node's scheduling algorithms and the forwarding policies implemented by the Quality of Service (QoS) framework. Specifically, the occurrence of microbursts and interface tail drops can be attributed to the node's inability to efficiently manage packet transmission rates, leading to QoS queue starvation and subsequent congestion. Furthermore, the Huawei Access Node's reliance on a centralized control plane, which is responsible for making forwarding decisions based on QoS policies, exacerbates the issue by introducing additional latency and processing overhead. The combination of these factors creates a complex interplay of control and data plane mechanisms that must be carefully navigated to resolve the congestion event.

Pre-requisite Checks

- Check: Verify the site's environmental conditions to ensure that temperature and humidity levels are within the recommended operating range for Huawei Access Nodes.

- Check: Confirm the status of redundant links and ensure that they are functioning correctly.

- Check: Verify that the node's power supply is stable and within the recommended operating range.

Potential Root Causes

Cause	Probability
Misconfigured QoS policies	High
Insufficient buffer memory allocation	Medium
Hardware interface congestion	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos policy, Expected Output: Policy Name Policy Type Policy Status, Description: This command displays the current QoS policy configuration, allowing the technician to verify that policies are correctly applied to the affected interfaces.

Command: display buffer memory, Expected Output: Memory Type Memory Size Memory Allocated Memory Free, Description: This command provides information on the node's buffer memory allocation, enabling the technician to identify potential issues related to insufficient memory allocation.
```

```
Command: display interface congestion, Expected Output: Interface Congestion Rate Buffer Size, Description: This command displays real-time information on interface congestion, allowing the technician to determine the extent of the issue and identify potential causes.
```

Resolution Steps

Step 1: Step: Verify and correct QoS policy configuration, Command: undo qos policy, Description: This step involves verifying the current QoS policy configuration and making any necessary corrections to ensure that policies are correctly applied to the affected interfaces.

Step 2: Step: Allocate additional buffer memory, Command: allocate buffer memory, Description: This step involves allocating additional buffer memory to the node to alleviate potential issues related to insufficient memory allocation.

Step 3: Step: Adjust hardware interface settings, Command: adjust interface congestion control, Description: This step involves adjusting the hardware interface settings to reduce congestion and improve overall network performance.

Post-Check Verification

```
Command: display qos policy, Expected Output: Policy Name Policy Type Policy Status, Description: This command is used to verify that the QoS policy configuration has been corrected and is functioning as expected.

Command: display buffer memory, Expected Output: Memory Type Memory Size Memory Allocated Memory Free, Description: This command is used to verify that the buffer memory allocation has been adjusted and is functioning as expected.

Command: display interface congestion, Expected Output: Interface Congestion Rate Buffer Size, Description: This command is used to verify that the interface congestion has been reduced and is functioning as expected.
```

Roll-back Procedures

- Step: Restore original QoS policy configuration, Command: apply qos policy, Description: This step involves restoring the original QoS policy configuration to reverse any changes made during the troubleshooting process.

- Step: Revert buffer memory allocation, Command: undo allocate buffer memory, Description: This step involves reverting the buffer memory allocation to its original state to reverse any changes made during the troubleshooting process.

- Step: Revert hardware interface settings, Command: undo adjust interface congestion control, Description: This step involves reverting the hardware interface settings to their original state to reverse any changes made during the troubleshooting process.